

Discussion 3 Linux Command Line vs the GUI

Discussion Topic:

The Linux operating system works mainly via the command line, with the Graphical User Interface (GUI) available for use, but not necessary.

Review the following video and, in your initial post, discuss the pros and cons of the command line vs the GUI:

McWilliams, G. (2024, August 29). The Linux command line [Video]. LinkedIn Learning.

Specifically, discuss how each of these environments help or hinder the ability to run multiple processes in parallel.

My Post:

Hello Class

When discussing the pros and cons of the command line vs the GUI, it is important to understand that they are not inherently competitors, but they are two different control layer systems for the Linux Operating System (OS). In his “The Linux command line” video, McWilliams (2024) explains that Linux does not require a GUI to interact with; the GUI is just a set of applications running on top of it, while the command line or terminal can interact with the system directly using a shell.

This direct access to the operating system through a shell has many advantages, including speed, efficiency, as it allows the user to wield substantial administrative power, and reduced system overhead as the console line is just one layer above the core kernel. This is important on systems where resources are restricted and executing services is the primary priority, such as servers or virtual machines. In this context, a GUI becomes an unnecessary burden that wastes critical computing resources.

The command line allows the user to run multiple processes; for example, Bourne-Again Shell (BASH) allows users to stop, resume, foreground, and background processes by controlling them in parallel from the same terminal (GNU Project, n.d.). This is powerful functionality for users like system administrators who need to run scripts, monitor logs, install packages, manage services, or run commands with long execution times and need to continue in parallel with other work or monitor other processes. However, the command line requires the user to know and remember commands, syntax, process IDs, job numbers, and shell behavior. McWilliams (2024) describes the command line as having a steep learning curve and is not ideal for graphical tasks such as browsing, video playback, or image editing. In other words, without a visual interface, a GUI, it can be easy to lose track of what is running, especially when several background processes write output to the same terminal.

The GUI also allows the user to run or monitor processes in parallel; this multitasking model is characterized by the ability to open multiple windows, switch between applications, monitor tasks with graphical tools, and manage several processes without the need to explicitly memorize each command. Terminal emulators such as GNOME Terminal combine approaches, command line / GUI, providing command line shell access inside the GUI that supports multiple tabs or profiles (Oracle, 2022). This

makes it easier to run several command-line sessions while still using graphical applications. The cons of a fully GUI OS management approach include its significant overhead, consuming more CPU and memory than a command-line-only system. Additionally, the GUI, by design, may hide technical details that are useful for a better understanding of the underlying architecture and for granular process control tasks.

In conclusion, the command line is better for automation, scripting, and for administrative process management & multitasking requiring precise controls. The GUI is better for visual multitasking and user-friendly interaction. Therefore, the best practical solution is not to treat one interface as superior in every use case. My personal preference is to use a hybrid approach like the GNOME terminal.

-Alex

References:

GNU Project. (n.d.). *Job control basics. Bash Reference Manual*.
https://www.gnu.org/software/bash/manual/html_node/Job-Control-Basics.html

McWilliams, G. (2024, August 29). *The Linux command line* [Video]. LinkedIn Learning.

Oracle. (2022). *gnome-terminal(1): User commands*.
https://docs.oracle.com/cd/E88353_01/html/E37839/gnome-terminal-1.html